Lecture 7

SM2501/FSG3084

Muscle biology and biomechanics
Part 1



1

But first… Project 1

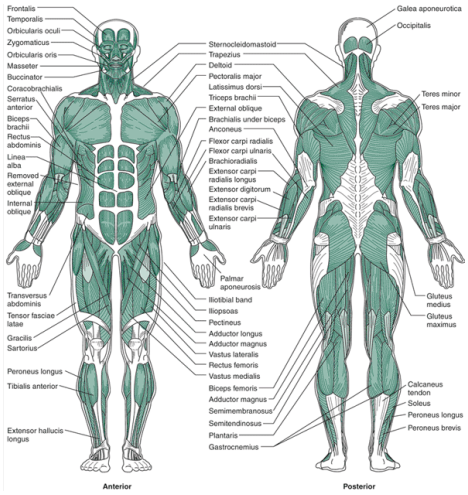
- Project deadline Friday 14 Nov. before 12.00
  - Upload to Canvas!
- From project instructions:

> About group work: The concept with group work is that the *group* works together on the project, not that each person does 1/3 of the project. That wouldn't be group work. Each person should understand the entire content and be able to answer some questions about all sections. Also, it is usually necessary to complete the first section to be able to complete the next section, so dividing up the different sections to different people will make it practically impossible for a person who hasn't been involved in the first section to do any sensible analysis for the second section, and so on, and the whole group's grade will likely suffer.
>
> In short, for group work, think *multiplication*, not *division*.

- Also: "Describe the contribution of each team member."
- Jennifer has office hours today at 13-15 at MoveAbility Lab, or otherwise by appointment.

2

Skeletal Muscle



Hamill & Knutzen 2009

3

Skeletal muscle

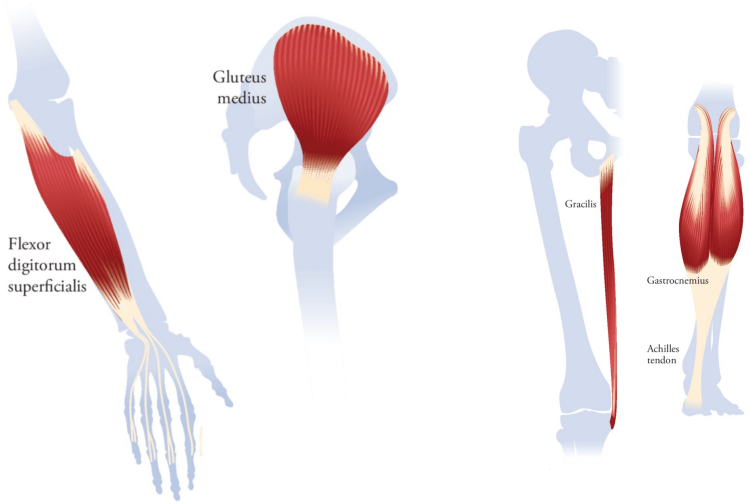
- Can only pull
- Forces applied to skeleton via tendons
- Very resilient tissue
- Performance determined by:
  - Excitability/Irritability: ability to response to stimulation
  - Contractility: ability to generation tension and shorten (shorten on average 57% resting length). *Unique to muscle tissue*
  - Extensibility: ability to lengthen or stretch
  - Elasticity: ability to return to resting length after stretch.

} *Connective tissues*



4

Attach to bone and span joints

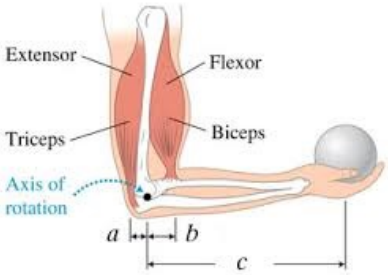


Architecture, attachments and geometry dictate function

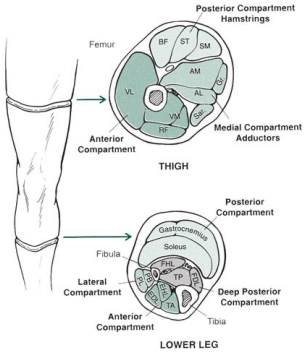
Uchida & Delp 2020

5

Described based on function/moment arm at joints







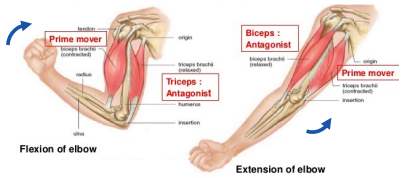
Muscles typically act in synergistic groups, not individually
- "~~quadriceps moment~~" -> "knee extensor moment"

Hamill & Knutzen 2009

6

Synergists: Agonists/prime movers & antagonists

- Specific to the desired motion; prime mover(s) or agonist(s) performs the intended motion
- Any muscle that opposes the action of the prome mover is an antagonist
- E.g. biceps brachii opposes the action of the triceps brachii to extend the elbow joint. The reverse is true to flex the elbow joint.
- Net moment can be zero
  - if there is no muscle contraction in either agonists or antagonists
  - in co-contraction – if e.g. flexor moment and extensor moment are in balance



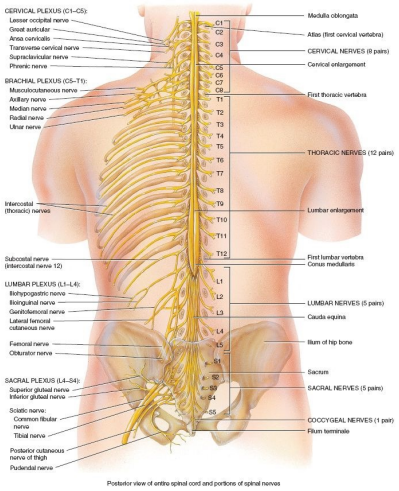
7

Muscle tissue

- Types:
  - striated (striped)
    - > Cardiac (not under voluntary control)
    - > Skeletal (under direct voluntary control)
  - non-striated (smooth)
    - > Not under voluntary control
    - > Ex. Veins, arteries, bladder, iris of eye

8

Neural pathways from CNS to muscles



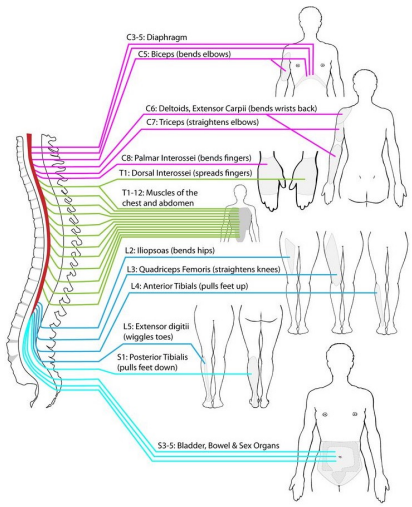
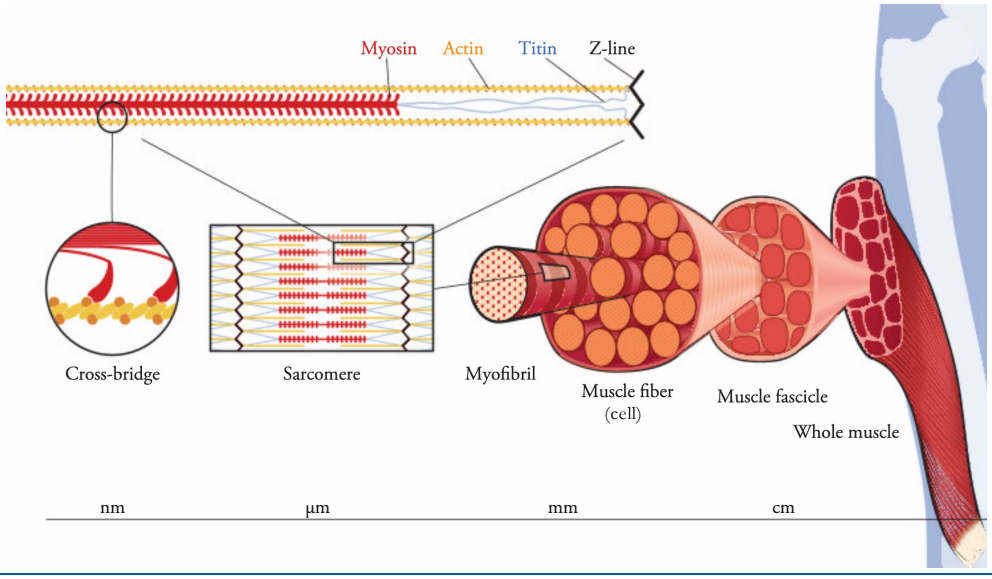


Image by Tamityville via Flickr

9

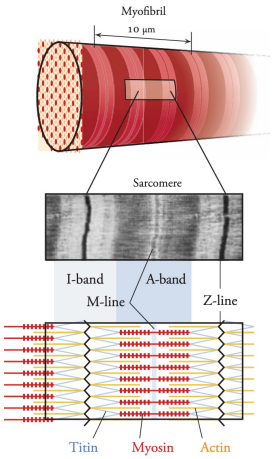
Multiscale structure of muscle



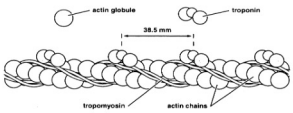
Delp & Uchida 2020

10

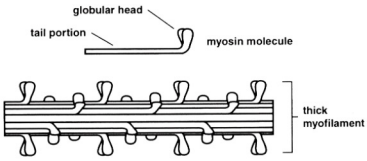
Sarcomeres: Contractile unit of muscle

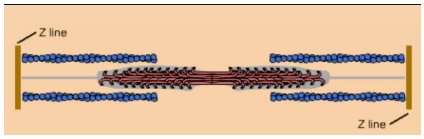


Actin (thin filament)



Myosin (thick filament)



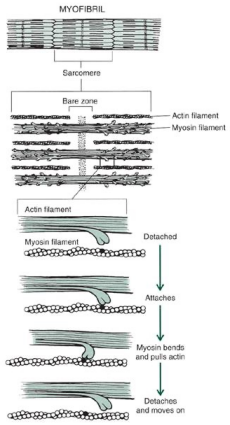


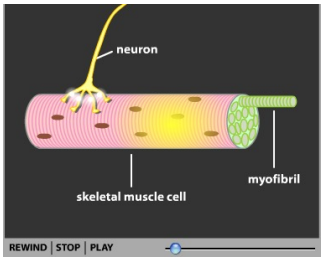
Myosin binds to actin in the presence of calcium ions (action potential)

Nigg & Herzog 2005, Delp & Uchida 2020

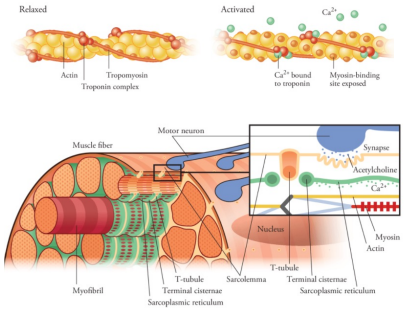
11

Activation and Sliding Filament Theory





http://www.sumanasinc.com/webcontent/animations/content/muscle.html
Garland Science Publishing

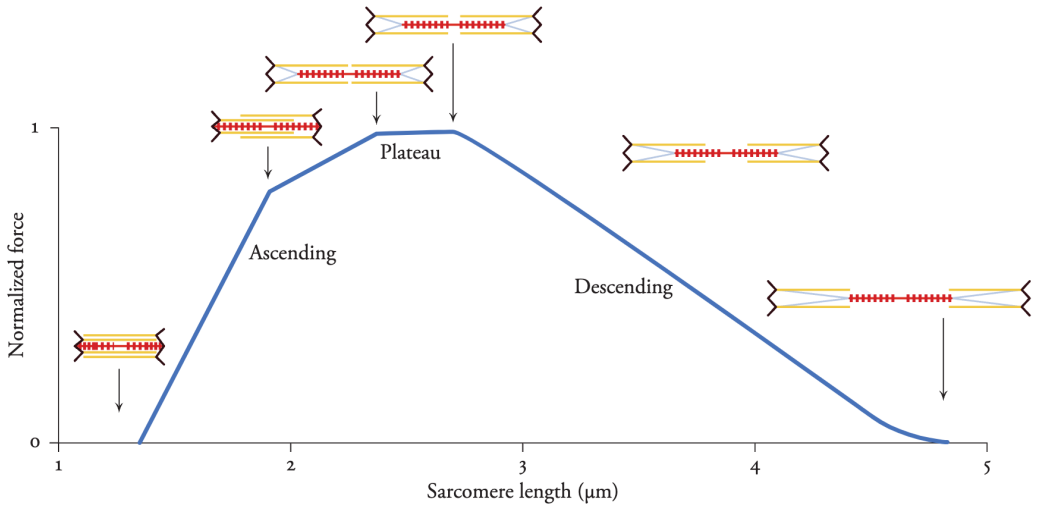


Hamill & Knutzen 2009

Delp & Uchida 2020

12

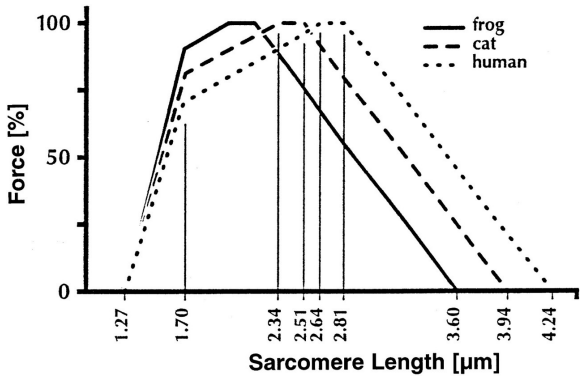
Sarcomere active force-length relationship



Delp & Uchida 2020

13

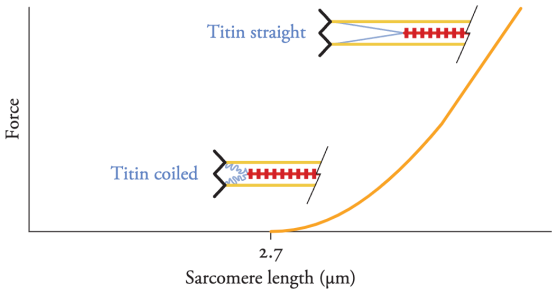
Sarcomere optimal length



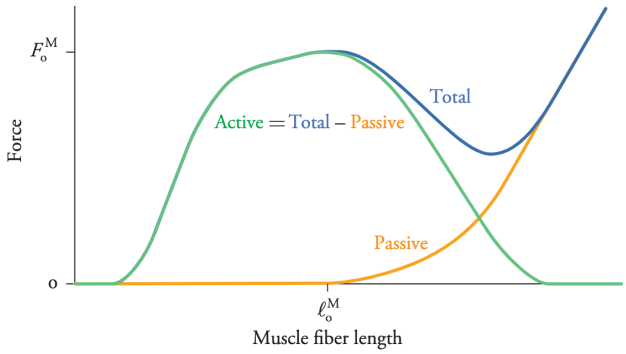
14

Force-length relationship (active + passive)

Sarcomere passive F-ℓ relationship



Muscle fiber F-ℓ relationship

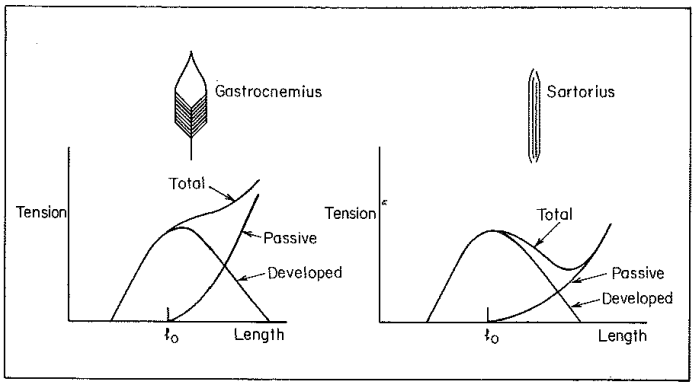


$F_O{}^M$ peak active force at $\ell_O{}^M$ optimum muscle fiber length

Delp & Uchida 2020

15

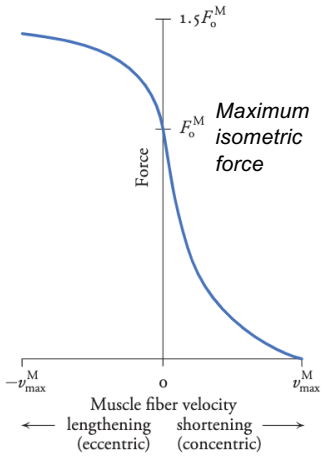
Force-length relationship varies between muscles

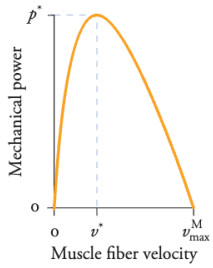


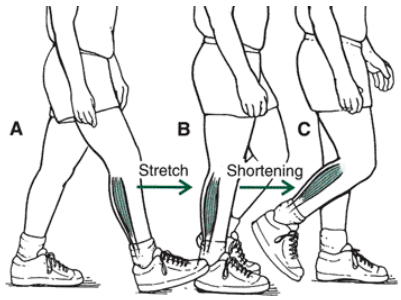
McMahon 1984

16

Force-velocity relationship







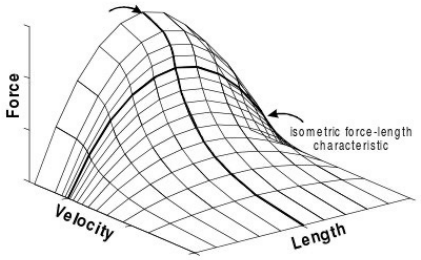
Delp & Uchida 2020

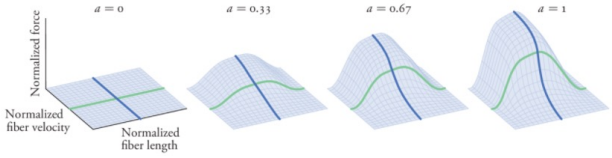
Hamill & Knutzen 2009

17

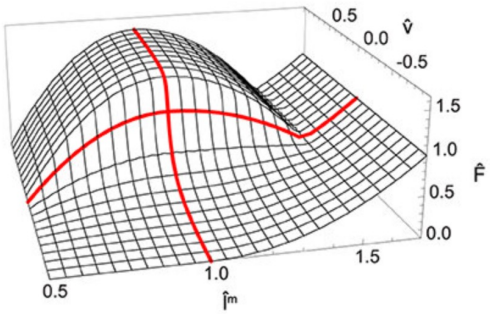
Contraction dynamics muscle *F-l-v* relationship

Active F-ℓ-v relationship





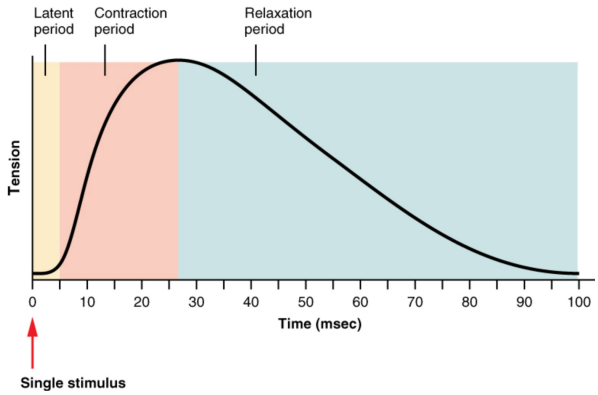
Active+passive F-ℓ-v relationship at full activation



Sabatelli 2015, Uchida & Delp 2020, Buhrmann & Di Paulo 2014

18

Activation/Twitch and Force Production

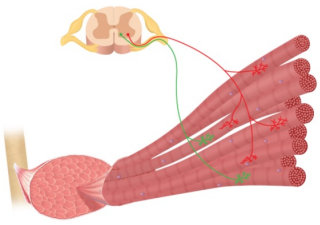


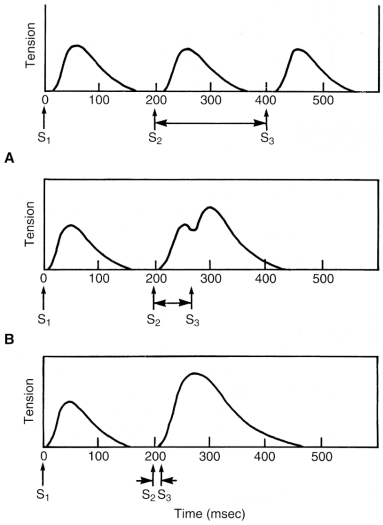
- Latent period: 1-2 ms, Ca^{2+} diffusion, formation of cross bridges
- Contraction period: cross bridge cycling
- Relaxation period: Ca^{2+} reabsorbed, muscle returns to normal length

Twitch duration varies between 10-100 ms for different muscles

19

Muscle stimulus (twitch)

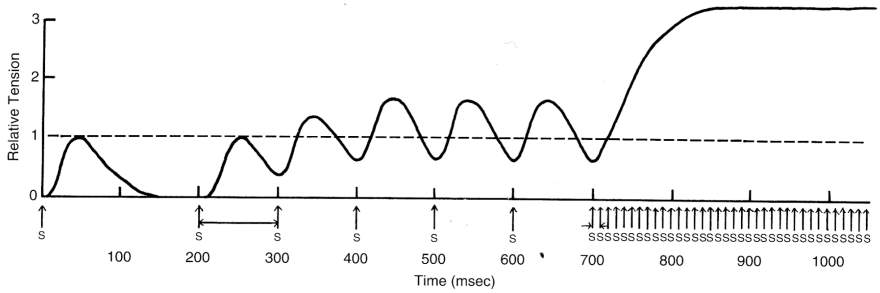




Hamill & Knutzen 2009

20

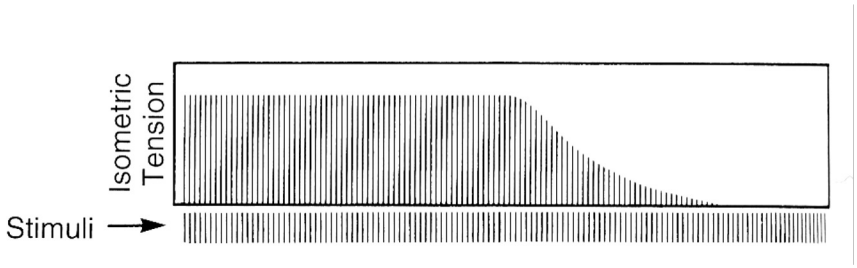
Tetanus



Hamill & Knutzen 2009

21

Fatigue

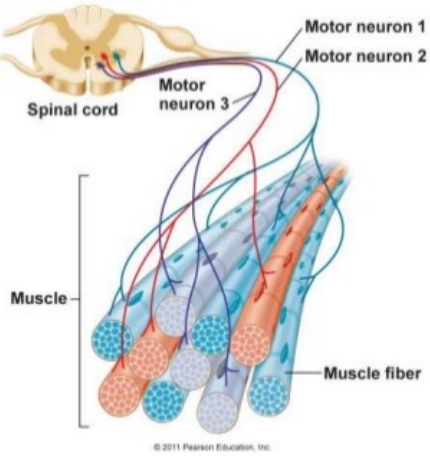


Hamill & Knutzen 2009

22

Motor unit

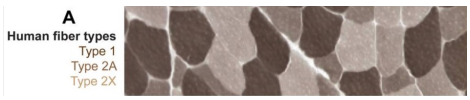
- A single motor neuron with all the muscle fibers innervated by it constitutes a motor unit
- There are 3 types of motor units – slow/Type I, fast fatigue-resistant/Type IIa, fast fatigable/Type IIx



23

Fiber Types: myosin heavy chain expression

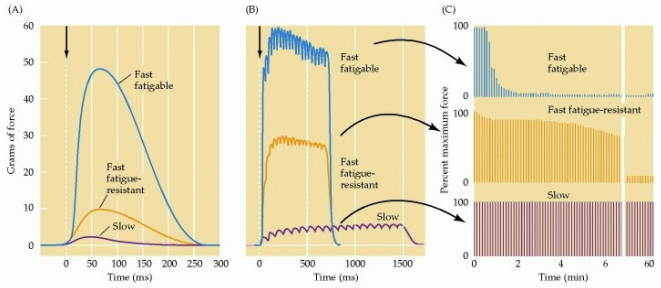
- Type I: Slow-twitch oxidative (SO)
  - Produce ATP aerobically
  - Rich in capillaries, mitochondria and myoglobin (red color)
  - Slow contraction time (>70 ms)
  - Relatively small diameter, relatively little tension
  - Slow to fatigue
- Type IIa: Fast-twitch oxidative (FO)
  - Produce ATP aerobically
  - Rich in mitochondria but not in myoglobin (lighter than SO)
  - Relatively fast contraction (30–50 ms)
  - Will eventually fatigue
- Type IIx: Fast-twitch glycolytic (FG)
  - Produce ATP via anaerobic glycolysis
  - Few mitochondria and little myoglobin – (light/white color)
  - Fast contraction (30–40 ms)
  - Usually large diameter, produce great tension for a short time
  - Fast to fatigue
- (Type IIb: not in humans. Largest, fastest and quickest to fatigue)
- Average population: 50-55% I, 30-35% IIA, 15% IIx. Number and size of Type II fibers decrease with age.
- Varies greatly among muscles & individuals (Type I in quads: 20% in sprinters -> 95% in marathon runners)



Talbot & Maves, 2017

24

Force and fatiguability of motor units types

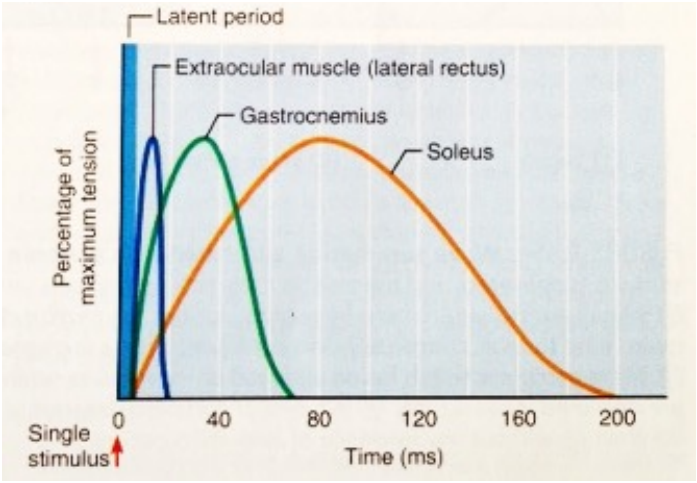


A. Tension in response to a single motor neuron action potential
B. Tension in response to repetitive stimulation of motor neurons
C. Response to repeated stimulation at a level that evokes max tension

Purves, Augustine et al, Neuroscience 2nd Ed, 2001

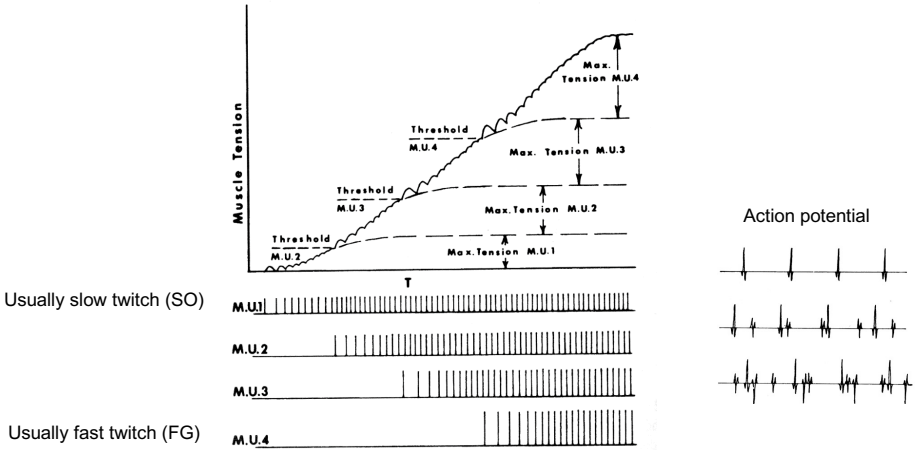
25

Twitch & force production in different muscles



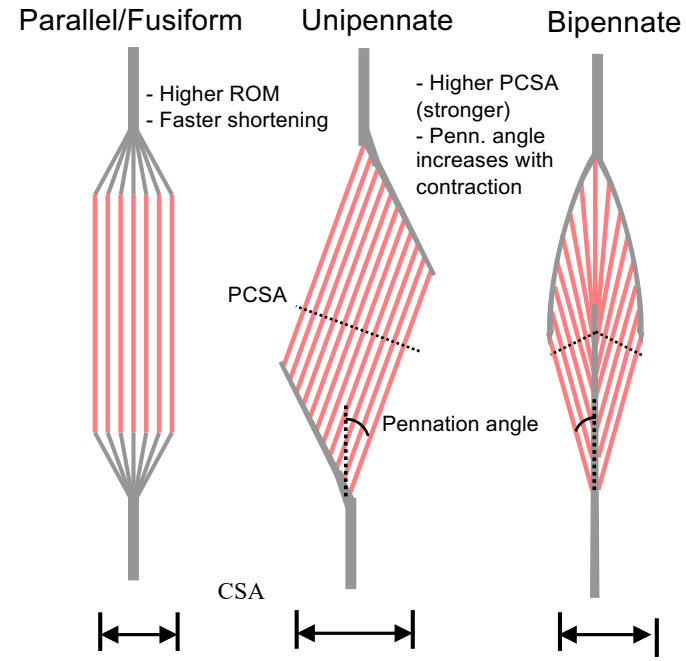
26

Recruitment of motor units

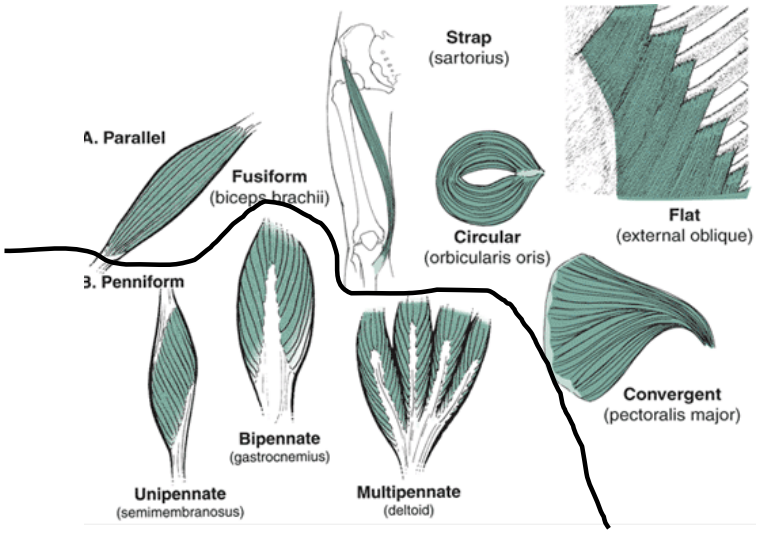


Winter 2005

27



28

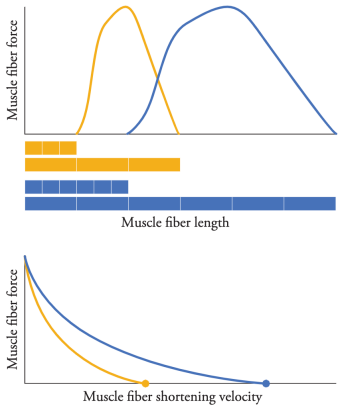


Hamill & Knutzen 2009

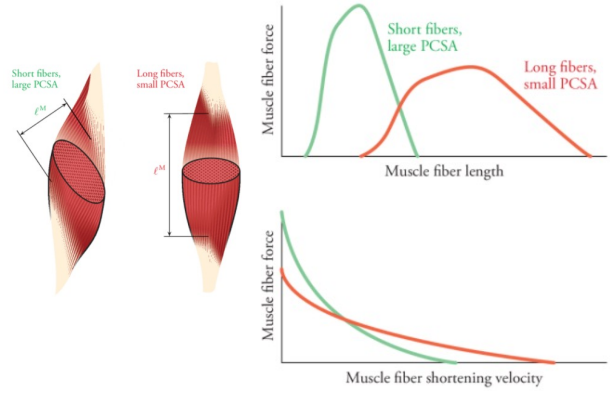
29

Sarcomere organization and fiber force

Muscle fiber force



Sarcomere distribution and fiber force



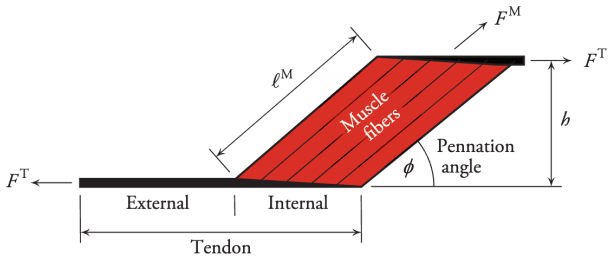
…All else being equal, a pennated muscle will be able to generate greater force than a parallel-fibered muscle of the same volume, but over a smaller range of lengths and at lower contraction velocities. -S. Delp

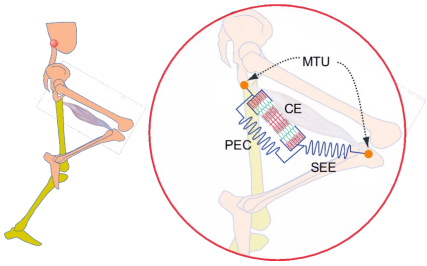
Uchida & Delp 2020

30

31/10/25

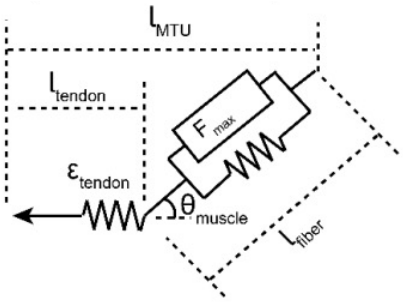
Muscle-*tendon* unit (MTU)
(or muscle-tendon actuator)





Series Elastic Element
Parallel Elastic Component
Contractile Element

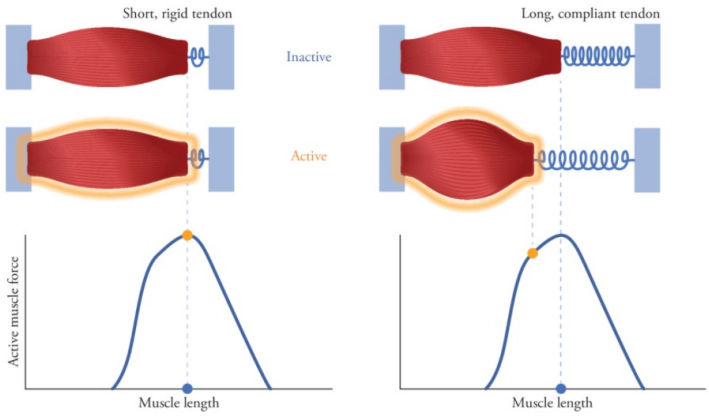
Hill-type muscle model



31

31/10/25

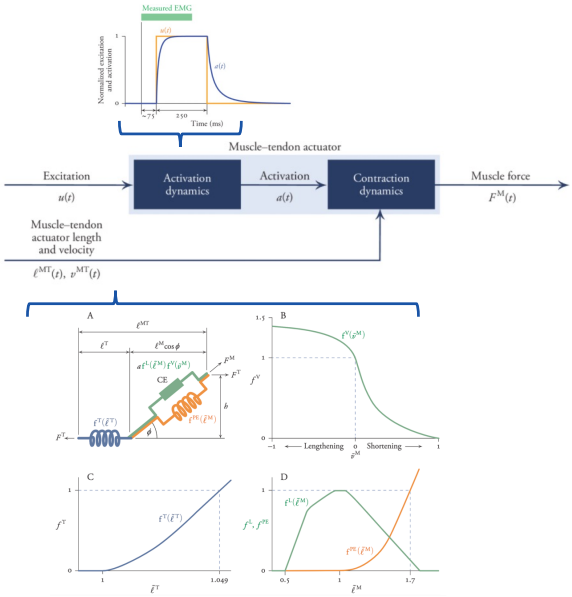
Tendon length and compliance in MTU



Uchida & Delp 2020

32

Excitation -> MTU force: putting it together



Uchida & Delp 2020